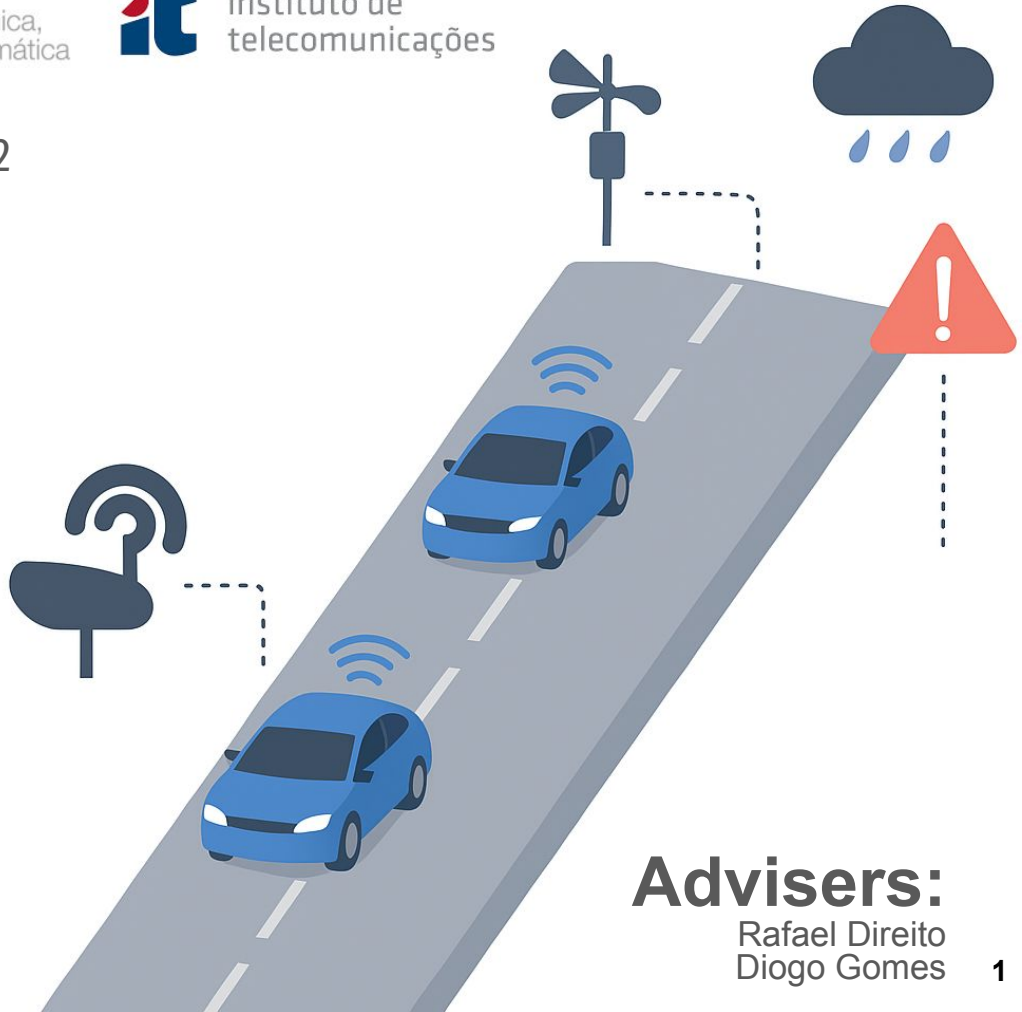
deti universidade de aveiro
departamento de electrónica, telecomunicações e informática

instituto de telecomunicações

Projeto em Engenharia Informática - MS2.2

Automotive App



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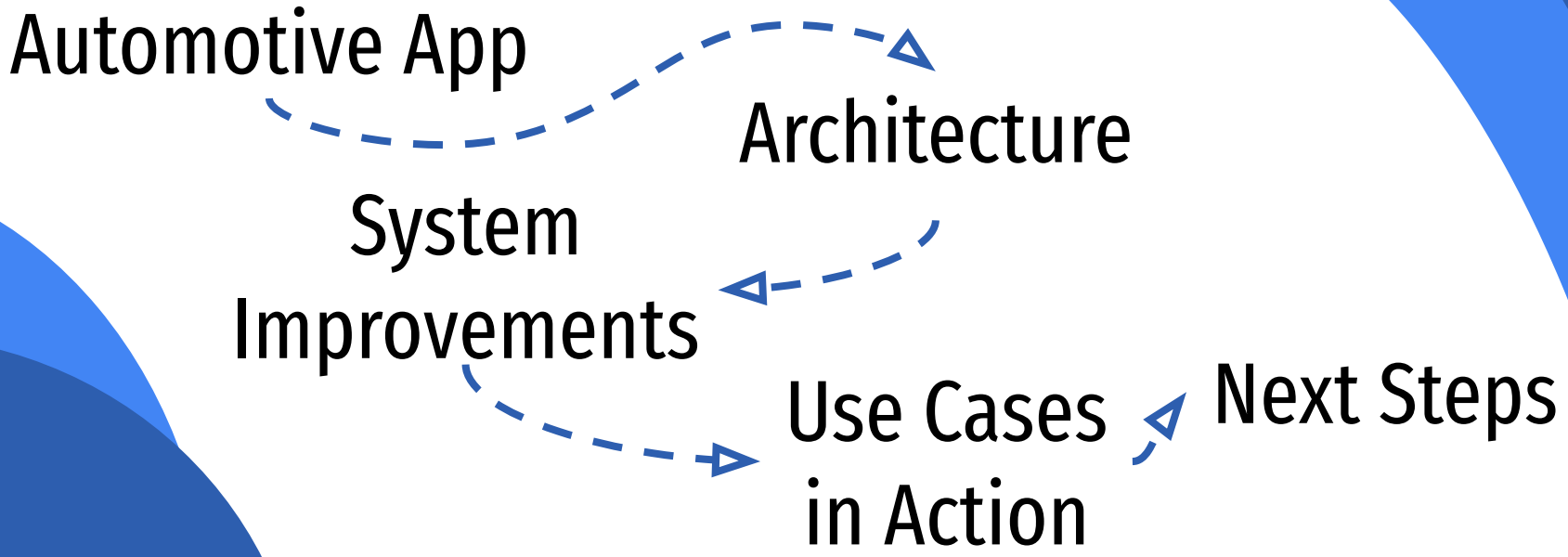
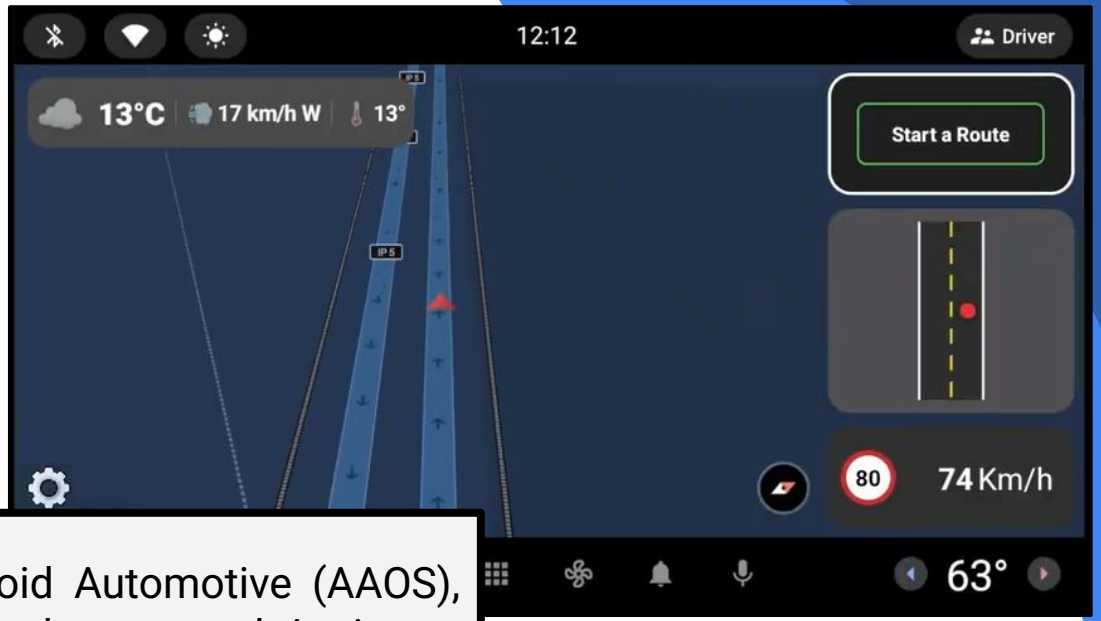
Prototype


Automotive App
Architecture
System Improvements
Use Cases in Action
Next Steps

Context

Automotive App



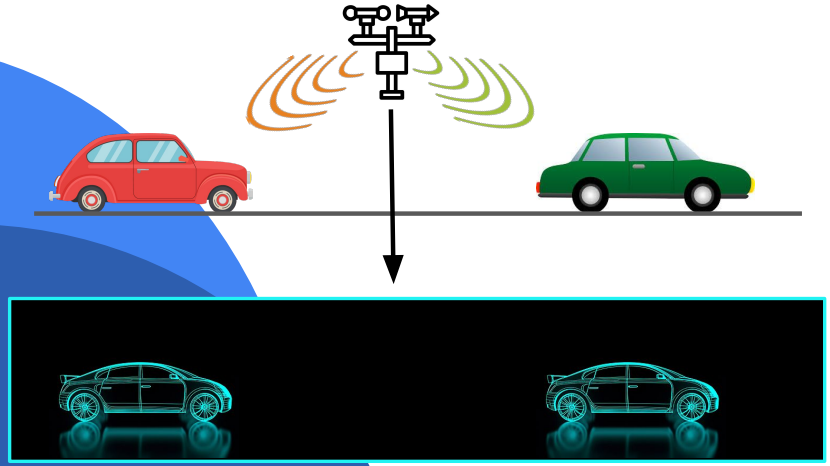
Objective:

Develop of an app for Android Automotive (AAOS), capable of leveraging sensor data around Aveiro to provide an Automotive Unified Road Assistant.

Oportunity

Digital Twins for Mobility

Claude2Edge infrastructure, orchestrated with Kubernetes



DT4MOB infrastructure provides:

- Faster communication and interaction between objects
- Rapid and efficient event detection
  (perfect for a road assistance app)

Automotive App can:

Provide knowledge of:

- Speeding
- Traffic Congestions
- Overtakes
- Emergency Vehicles
- Accidents
- Weather Conditions

Assist with:

- Maneuver Coordination
- Navigation

Risks

Information needs to be accurate and can not be too distracting, otherwise can be harmful to drivers.

Sensors can be faulty or break, compromising the entire system.

Mitigations:

Non Functional Requirements

- **Performance**: Alerts displayed ≤ 2s
- **Accuracy**: GPS error ≤ 5m
- **Scalability**: 100 000 users supported
- **Reliability**: Works with 9s disconnection

Actors

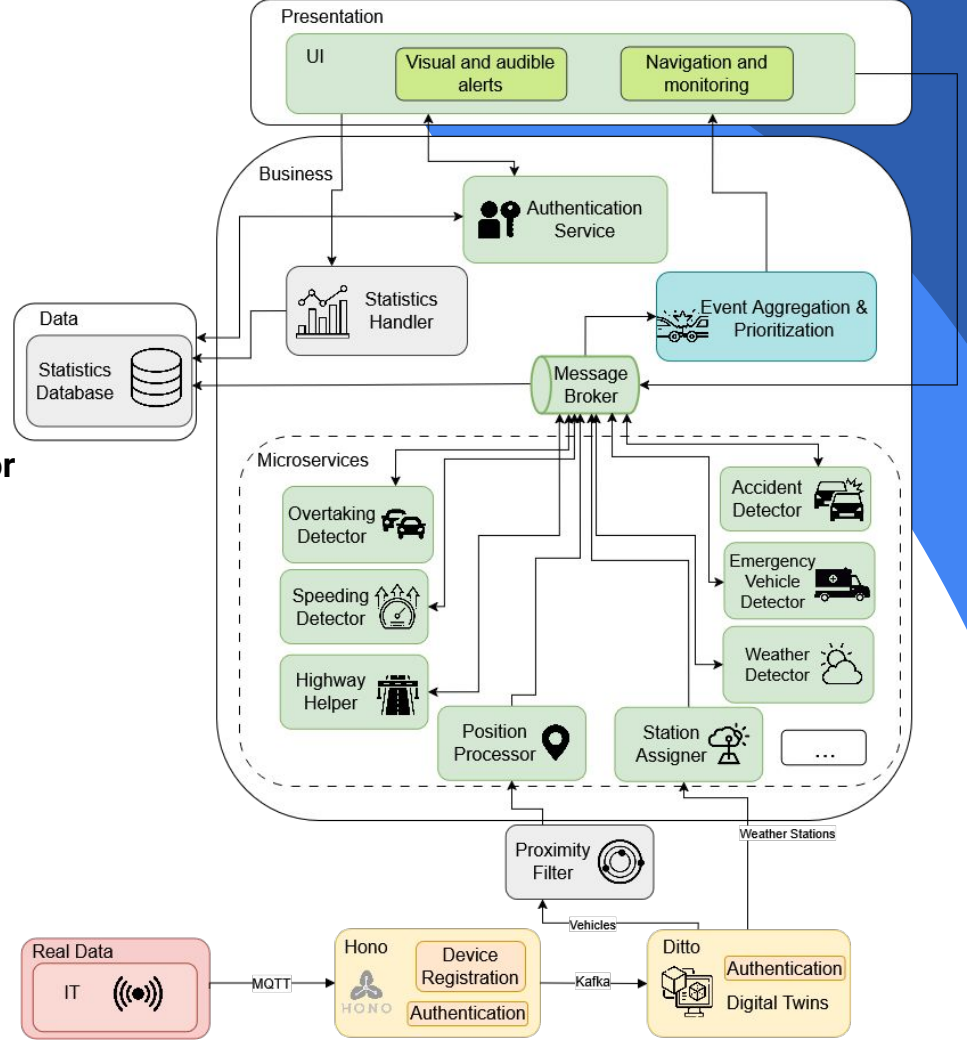
- Daily Driver
- Professional Driver
- Beginner Driver

- Relevant Information
- Simple, Clean and Intuitive Interface
- Customizable and Accessible Alerts

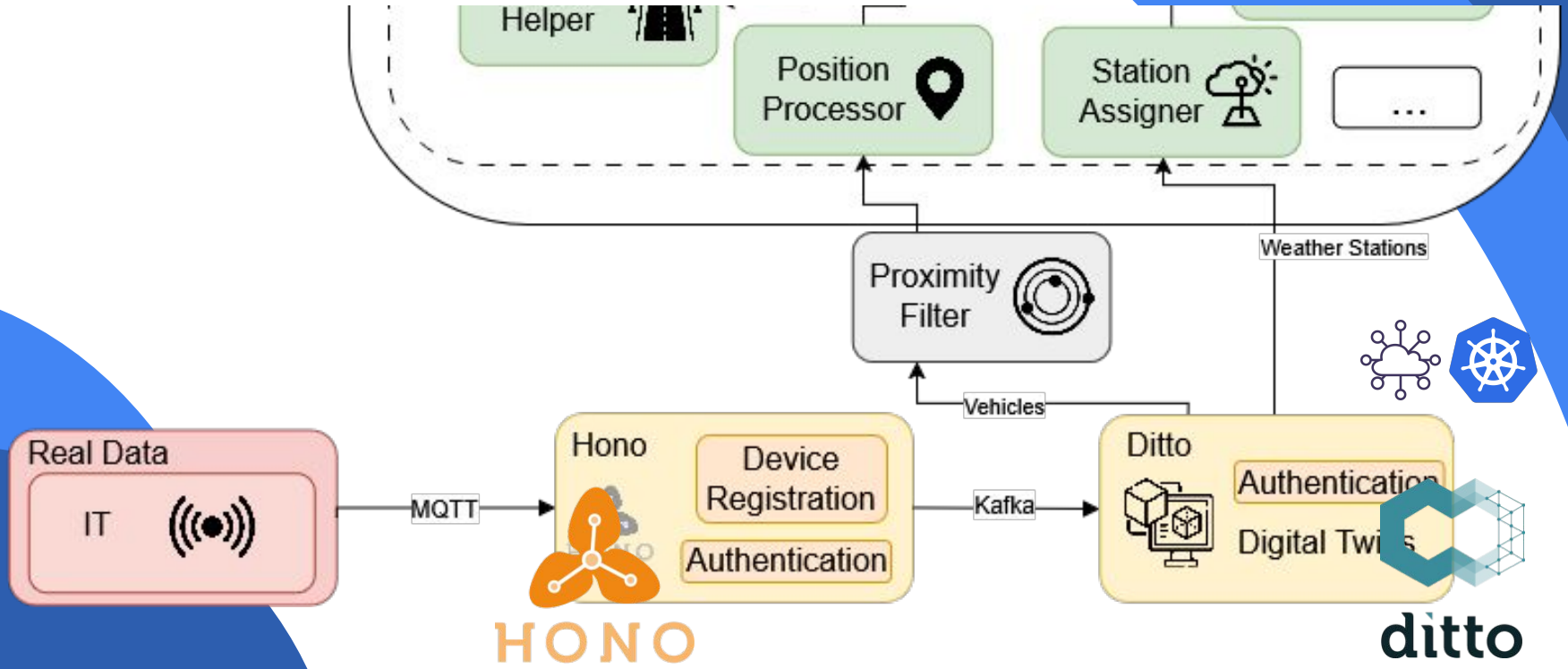
Architecture

Full Architecture

- Fully developed for this project
- Used for validation, out of scope for final solution
- Out of the scope of this project
- To be developed in the future
- Finalizing Development

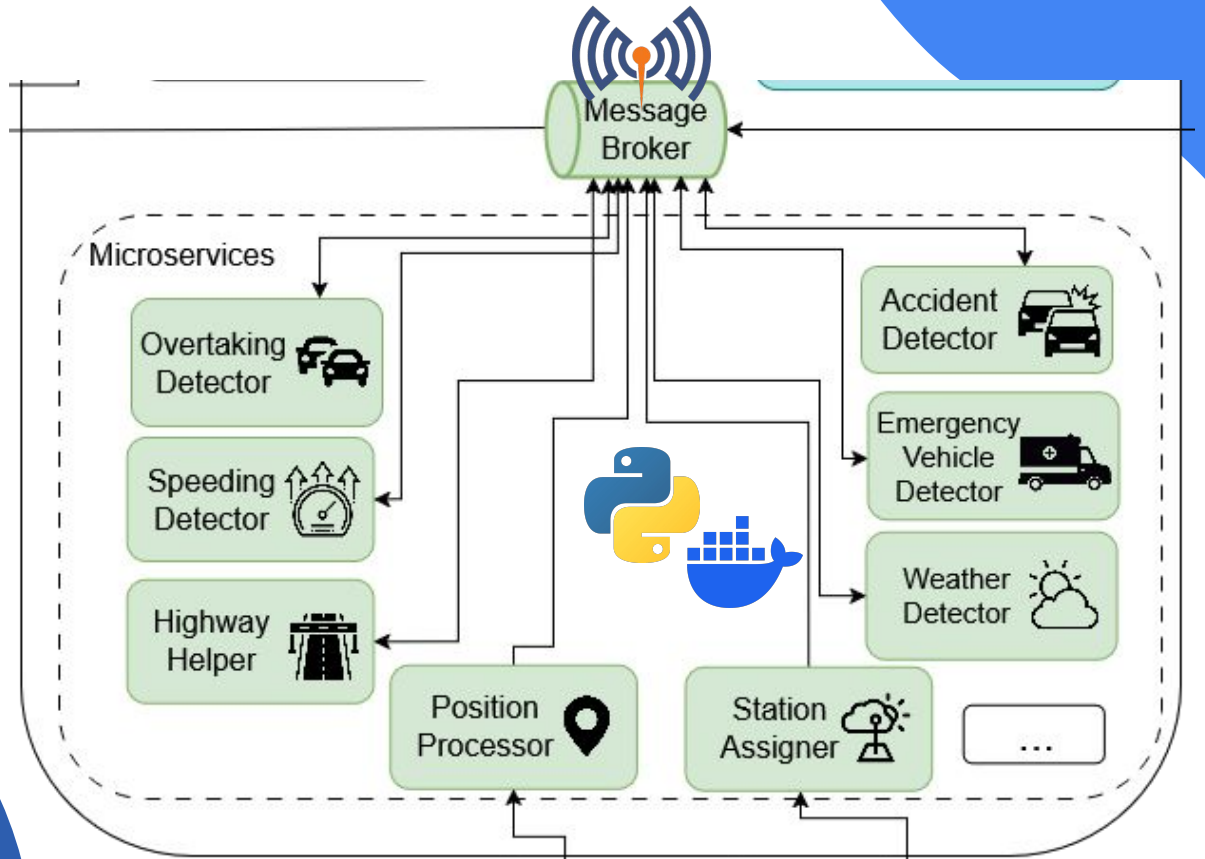


Digital Twin Processing

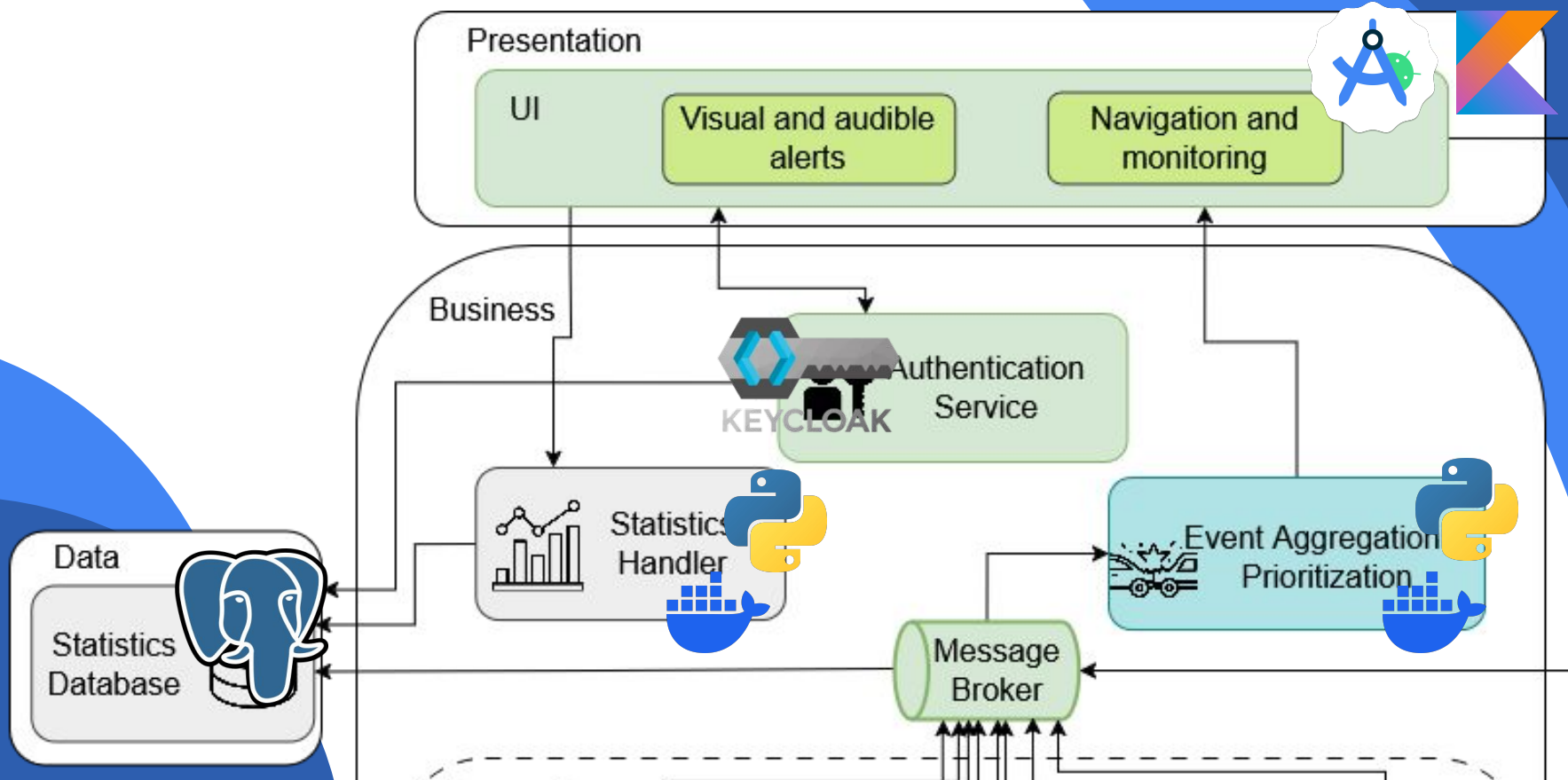


Proximity Filter: Concept

Event Detection

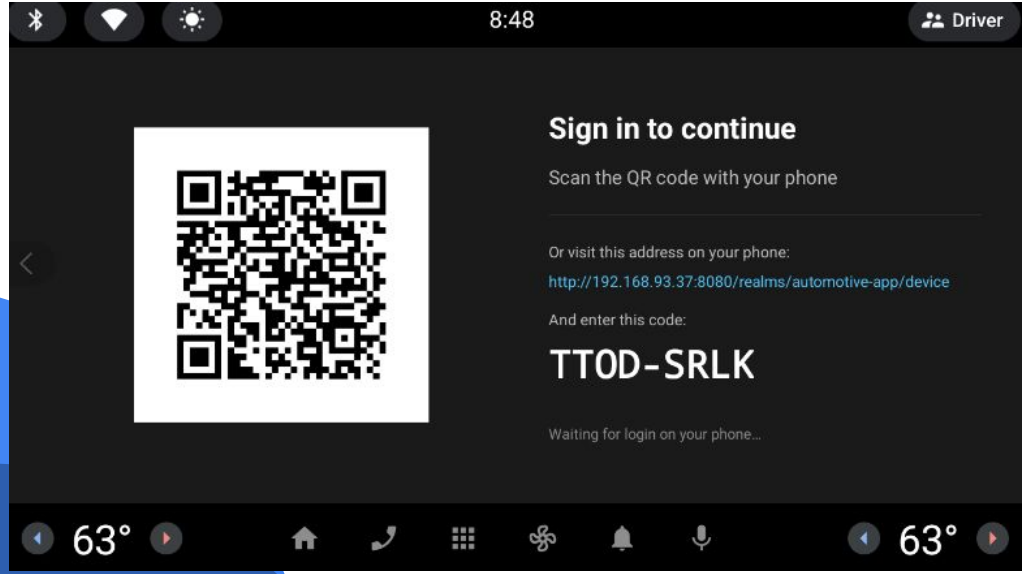


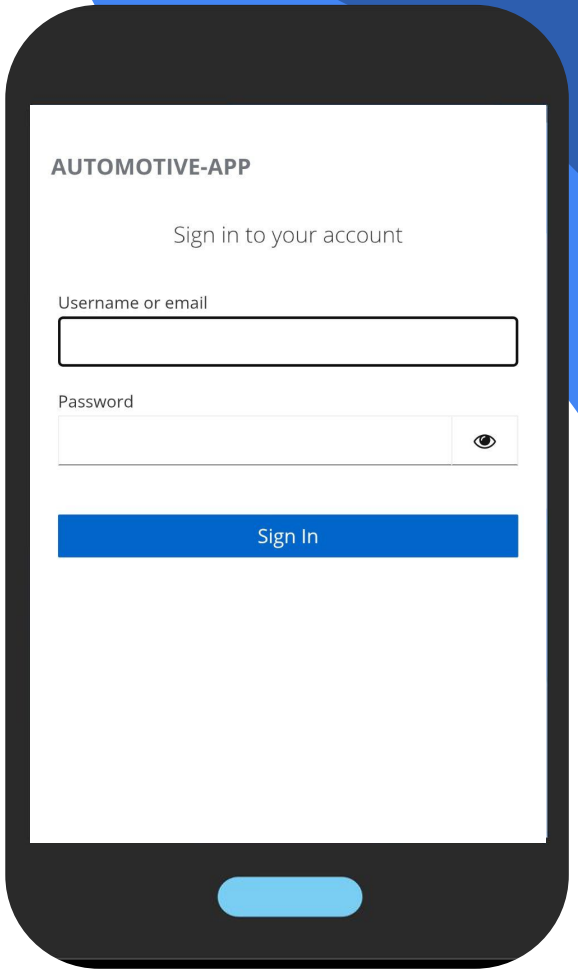
Frontend Interaction



System Improvements

Login System



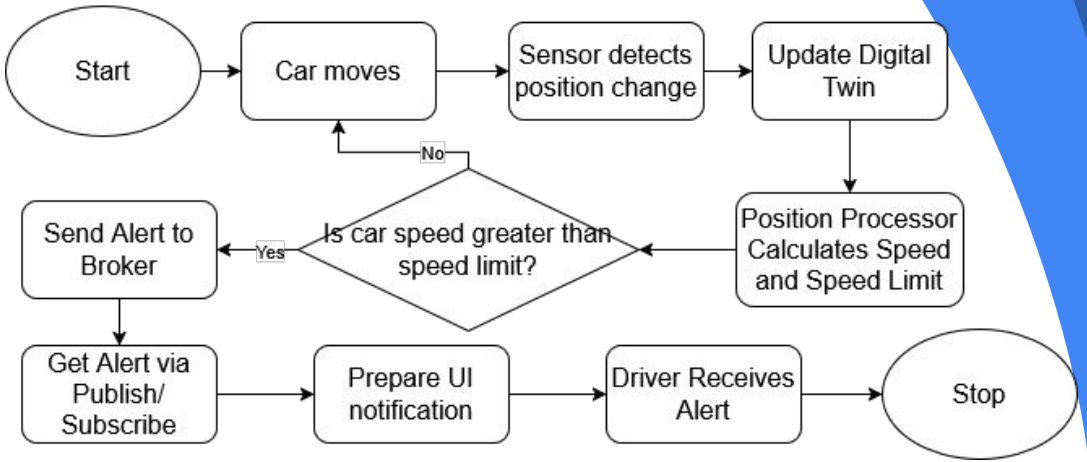


Use Cases in Action

How do we validate our UCs?

Recreate DT infrastructure with simulated data of each use case, created by ourselves.

System flow: Speeding Detection



Why Simulations?

Validating specific use cases with real data is unreliable (and dangerous)

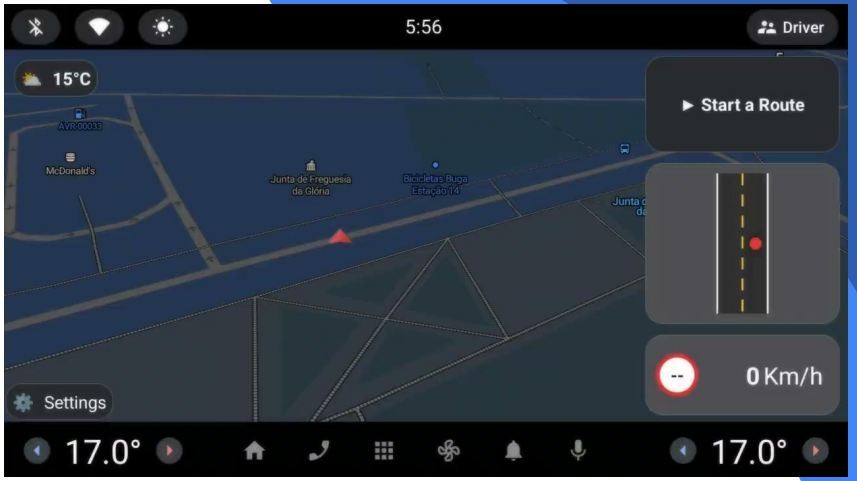
How can we test “real” UCs?

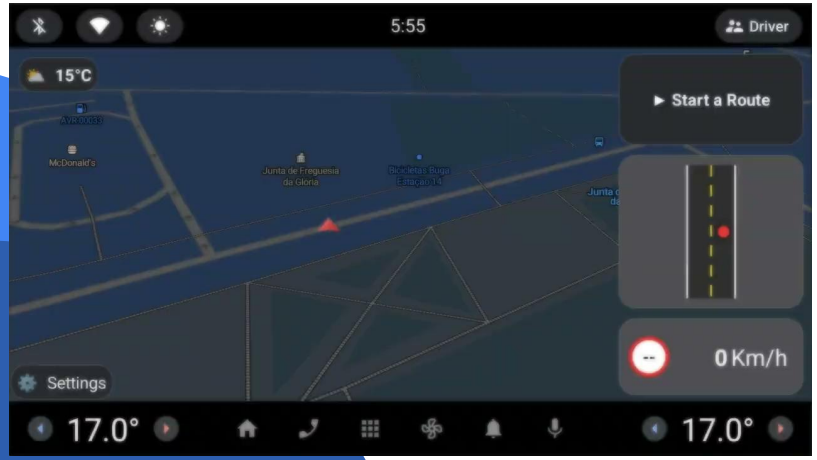
- Run our app on an Automotive OS system (Khadas VIM3 Pro) with an Automotive Screen.
- Drive around Aveiro and record relevant events.

Use Cases in Action

View Overtaking Alerts

The driver is notified when another vehicle is overtaking, increasing their awareness and safety.





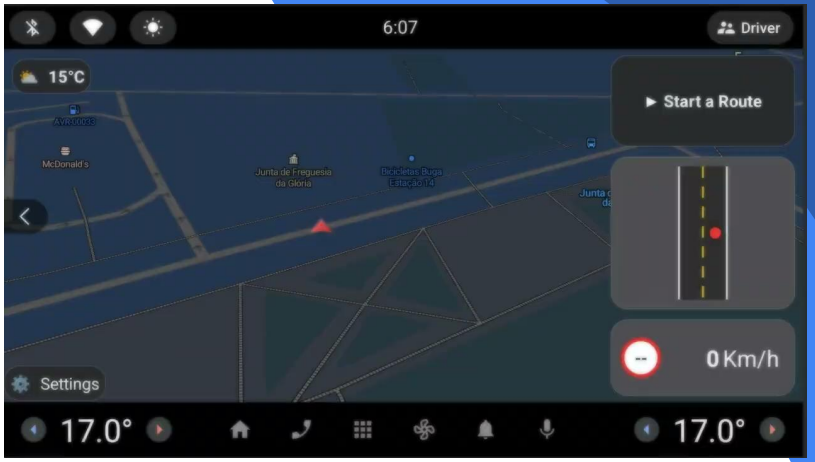
Monitor Speed Limit and Current Speed

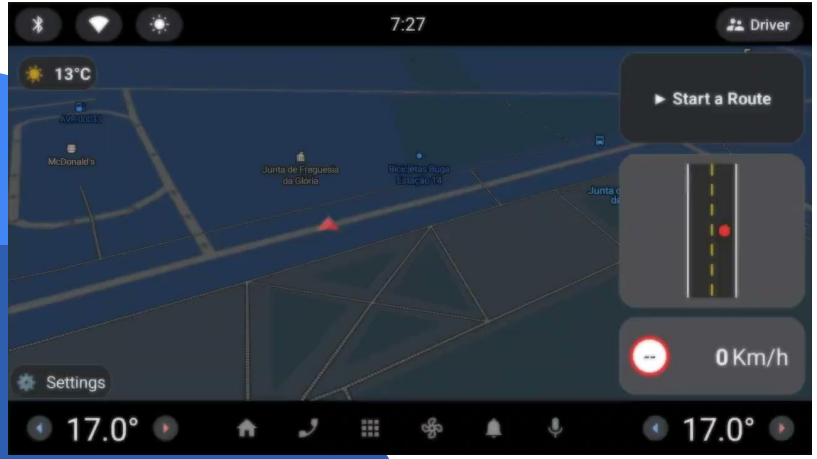
The driver is alerted when the vehicle exceeds the speed limit, as the system detects the speed limit of the road while monitoring current speed.

Use Cases in Action

Be aware of Accidents Ahead

The Driver is notified when a registered accident is located ahead, increasing their awareness and safety.





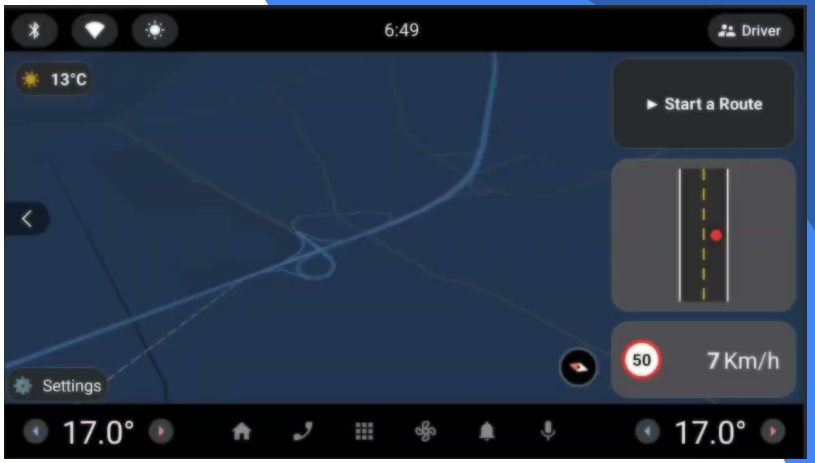
Identify Traffic Congestions

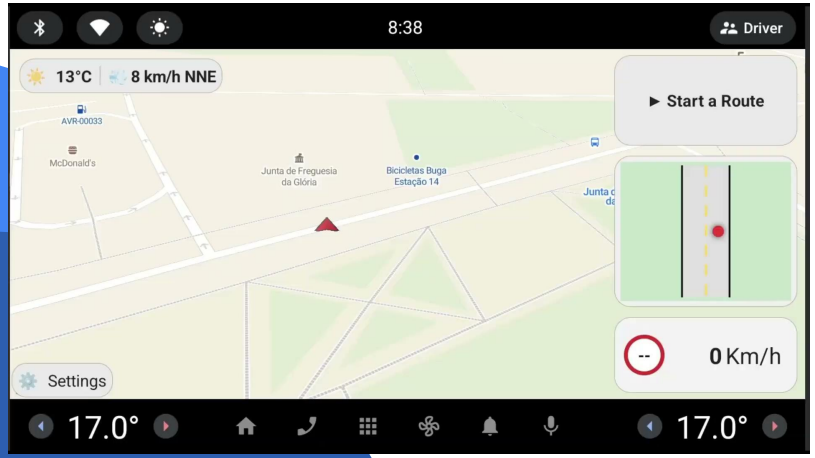
The Driver if informed on forwards traffic congestions, and can knowingly change their route to optimize travel time.

Use Cases in Action

Enter highways (unsafe)

The Driver receives a warning when is unsafe to enter a highway.





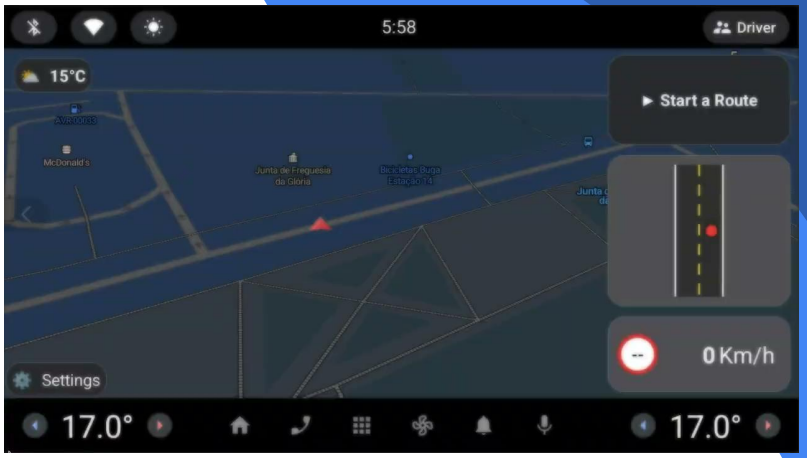
Enter highways (safe)

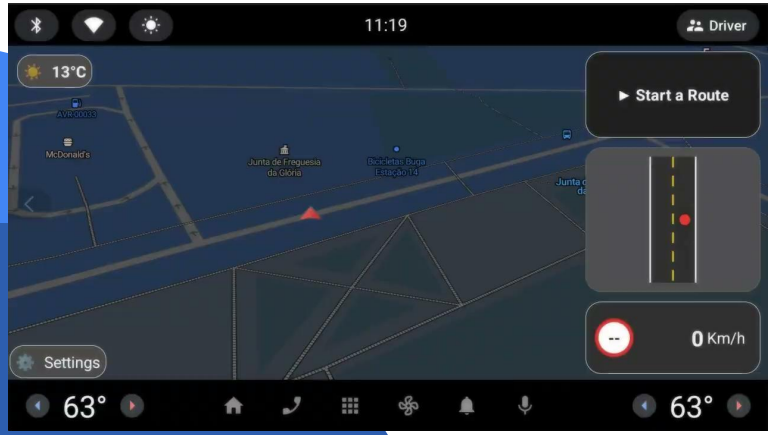
The Driver receives is notified when is safe to enter a highway.

Use Cases in Action

Be alerted when an emergency vehicle approaches

The Driver can be aware and give way to an emergency vehicle on their vicinity.





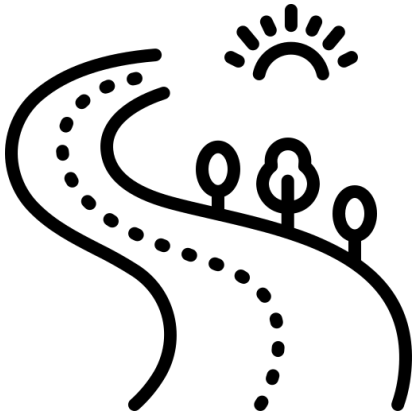
Know detailed weather information

The Driver can have access to detailed and customisable information on weather conditions, as well as be notified on extreme conditions, detected by their nearest weather station, so they can more easily monitor vehicles health and drive safer when needed.

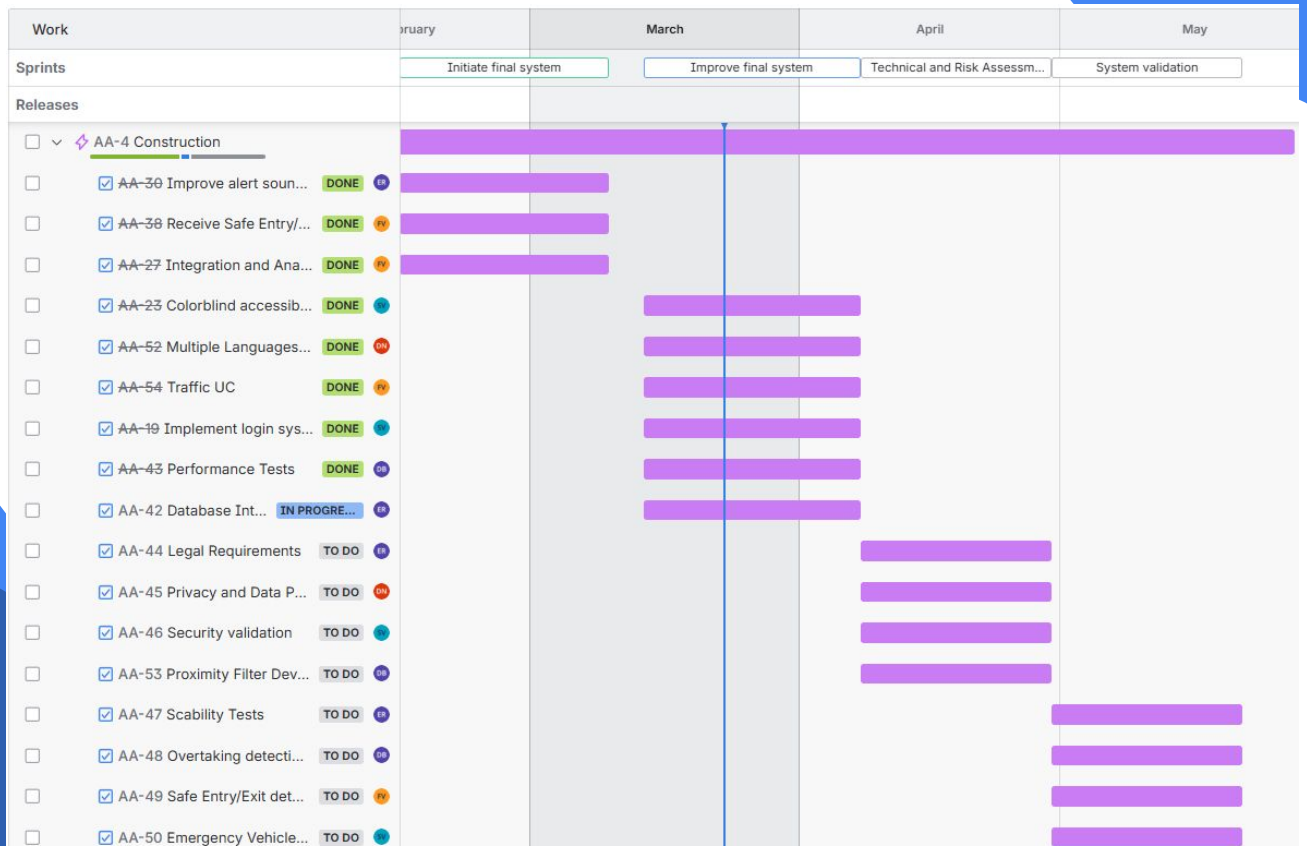
Next Steps

Next Steps

- Finish Event Prioritization
- Finish Database Implementation
- Refine and Record Performance (Proximity Filter)
- Security Improvements
- Match Legal Requirements



Next Steps - Calendar



Find Out More

Microsite

https://atnog.github.io/pei-automotive-microsite/

Jira

http://ua-pei-automotive.atlassian.net/